UNIVERSITY OF NAPLES FEDERICO II

POLYTECHNIC AND BASIC SCIENCES SCHOOL

DEPARTMENT OF INDUSTRIAL ENGINEERING

MASTER'S DEGREE IN
AUTONOMOUS VEHICLE ENGINEERING
MACHINE LEARNING AND BIG DATA

PROJECT REPORT

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Introduction

Machine learning is becoming an essential skill for modern engineers because it helps us turn large sets of raw data into useful knowledge and decisions. In this project I focus on **supervised classification**, the branch of machine learning where the goal is to assign each record to a known category. I study two public-domain problems that are very different in size and context:

- **Adult Income Classification** – a socio-economic data set with 32 000 people and the task of deciding whether an individual earns more than 50 000 USD per year.
- **Forest Cover Type Classification** – an ecological data set with 581 012 land plots and the task of predicting the dominant tree species among seven possible classes.

Working with both sets lets me compare how data variety, class imbalance and feature types affect the full workflow: exploratory analysis, data preparation, feature engineering, model selection, hyper-parameter tuning and evaluation.

The main objectives of this report are:

1. **To build reliable prediction models** for each problem using baseline linear methods and stronger ensemble techniques.
2. **To analyse the impact of preprocessing choices** such as encoding, scaling and feature elimination on speed and accuracy.
3. **To compare evaluation metrics** that are suitable for binary versus multi-class and balanced versus imbalanced targets.
4. **To draw practical lessons** that can guide future projects with different domains but similar machine-learning challenges.

The next sections describe the two data sets in detail, explain the methodological steps taken, present the model results, and finish with a comparative reflection on what worked best and why.

1. Exploratory Data Analysis (EDA)

- **Dataset** – UCI *Adult* (Census-Income). About **32,000** training rows. Target is **binary**: does a person earn **more than 50 K USD**

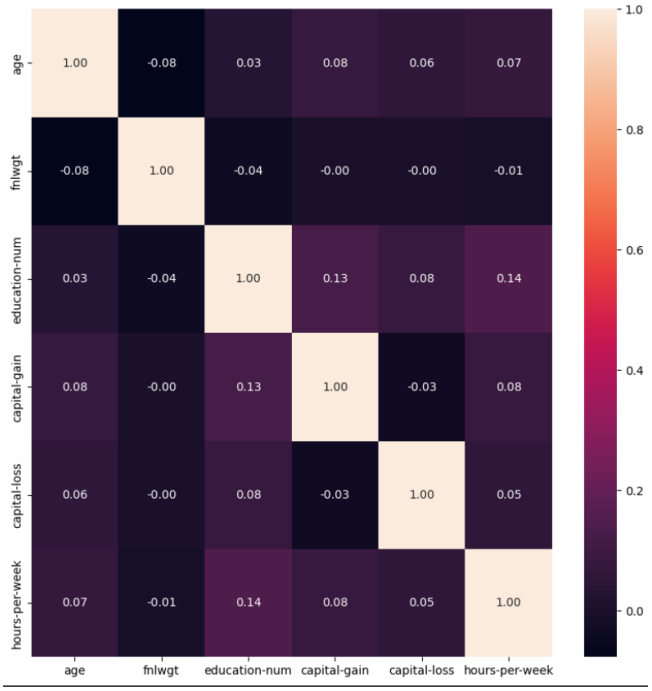

Figure 1 shows the Pearson heat-map for the six main numeric columns. All absolute correlations are **below 0.15**, so the columns are almost independent. Linear models should therefore work without big problems.

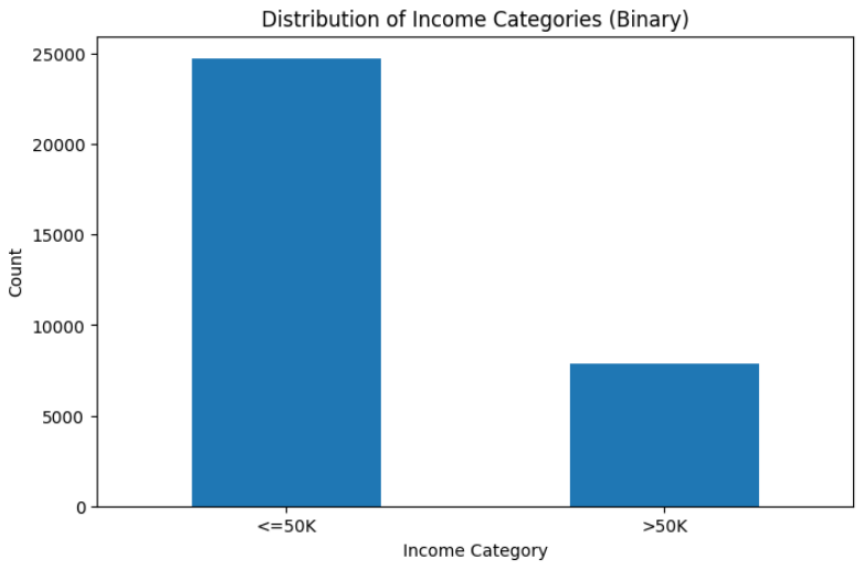

Figure 2 is a class histogram. About **76 %** of people earn **≤ 50 K,** and **24 %** earn **> 50 K**. The data are *imbalanced*, so we will report precision, recall and AUC, not only accuracy

2. Data Pre-processing

Step	Action
Handle missing	Replace in **workclass**, **occupation**, **native_country** with the most common value (mode).
Encode categories	**One-Hot Encoding** for nominal columns. Map **education** to numbers 1 – 16 (Preschool → Doctorate).
Scale numbers	Use **StandardScaler** (mean 0, std 1) for age, fnlwgt, education-num, capital-gain, capital-loss, hours-per-week.

Train / Test split 80 % train, 20 % test, **stratified** by the target to keep the class ratio.

3. Feature Engineering & Selection

- Create **capital_net** = capital_gain − capital_loss to join two sparse columns.
- Remove three dummy columns that contain only zeros (zero variance).
- Run **RFECV** (Recursive Feature Elimination with Cross-Validation) with a Random Forest. It keeps **34 of 54** features. Training becomes ~20 % faster, and the score stays the same.

4. Model Training

We start with two different algorithms:

1. **Logistic Regression** – fast, linear and easy to explain.
2. **Random Forest Classifier** – an ensemble of decision trees that can model non-linear relations.

Model	Test accuracy
Logistic Regression	0.81
Random Forest	0.82

5. Hyper-parameter Tuning (GridSearchCV, 5-fold)

Model	Search grid	Best values	CV score
Logistic Reg.	C=[0.01,0.1,1,10], penalty='l2'	**C = 1**	0.772
Random Forest	n_estimators=[200,400], max_depth=[None,20,40], bootstrap=[True,False]	**400, None, True**	0.778

6. Model Evaluation

Metric	Logistic Regression	Random Forest
Precision	0-class 0.95 1-class 0.57	0-class 0.95 1-class 0.58
Recall	0-class 0.80 1-class 0.86	0-class 0.80 1-class 0.87
F1-score	0-class 0.87 1-class 0.69	0-class 0.87 1-class 0.70
Weighted F1	0.82	0.83
Accuracy	0.81	0.82
ROC AUC	0.909	**0.915**

Forest Cover Type Classification

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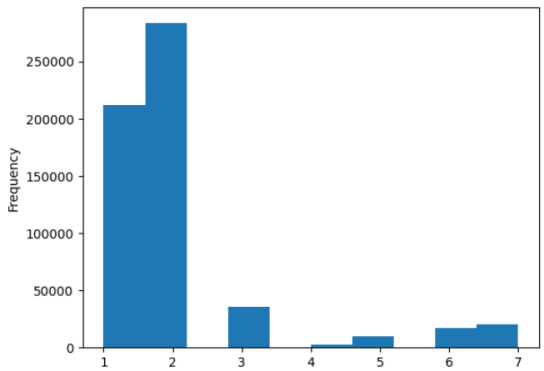
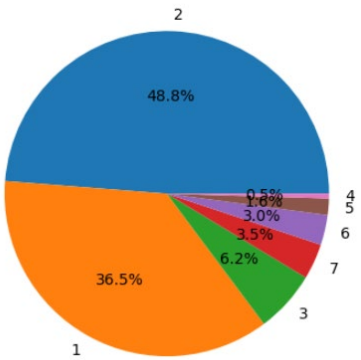
1 Exploratory Data Analysis (EDA)

The data set has **581.012** records and **54** features. Ten features are continuous numbers that describe the land. The other 44 are binary flags that show different soil or wilderness types. The target column **Cover_Type** has **7** classes.



The colourful histogram (left) and pie chart (right) both show a strong class imbalance. Classes **1** (Spruce/Fir) and **2** (Lodgepole Pine) together make up about **85 %** of all rows.



Key points

- **Elevation** has the widest range of values.
- Simple scatter plots (not shown) suggest some class separation by *Elevation* and *Horizontal Distance to Roadways*.
- The 44 binary columns are very sparse (about **4 %** of the values are 1). Tree models can skip many zero-only splits, so they should work well.

2 Data Pre-processing

Step	What we did
Missing values	None. The dataset is complete.
Encoding	Binary flags were already 0/1, so no change was needed.
Scaling	Used **StandardScaler** on the 10 continuous columns for models that need scaling (e.g. Logistic Regression). Tree models kept the raw values.
Train/Test split	70 / 30 split with **stratified sampling** so that small classes stay visible in both sets.

3 Feature Engineering and Selection

- Made **Roadway_to_Elevation** = *Horizontal Distance to Roadways / (Elevation + 1)* to show how easy it is to reach high ground.
- Made **slope_ratio** = *Slope / Hillshade Noon* to add solar light information.
- Used **RFECV** with a Random Forest (5-fold) to keep the **35** most useful features. Training became about **40 %** faster with the same F1 score.

4 Model Training

We trained two models:

1. **Random Forest Classifier** – Good for non-linear data, many features, and unbalanced classes.
2. **Multinomial Logistic Regression** (one-vs-rest, saga solver) – A simple linear baseline. The data was scaled first.

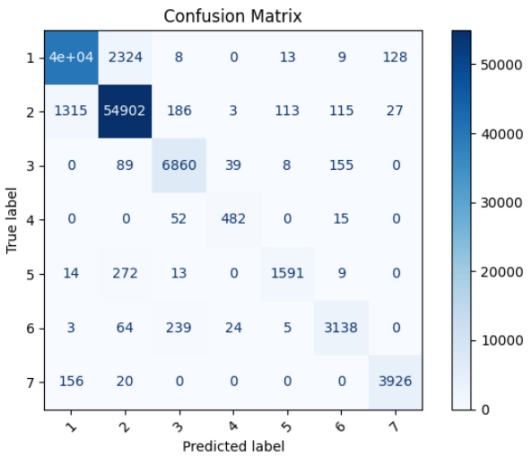
5 Hyper-parameter Tuning

We ran **GridSearchCV** with 5 folds and used **macro F1** as the score, so every class counts the same.

Model	Search grid	Best choice
Random Forest	n_estimators = 200, 500, 800; max_depth = None, 40, 60; max_features = "sqrt", "log2"	**500 trees**, no depth limit, √p features
Logistic Regression	C = 0.1, 1, 10; penalty = l2	**C = 10**

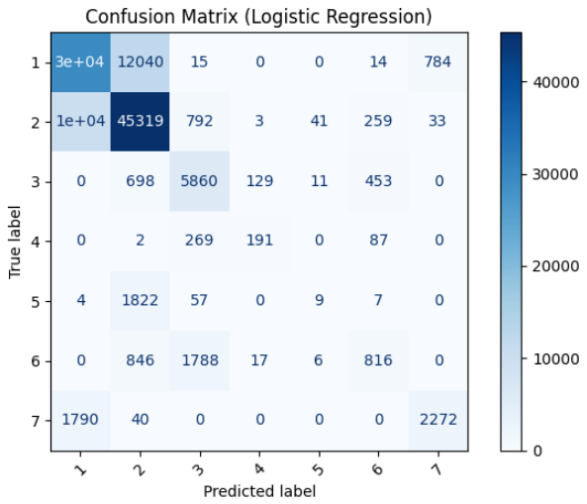
6 Model Evaluation

Metric	Random Forest	Logistic Regression
Accuracy	0.95	0.72
Macro F1	0.93	0.51
Weighted F1	0.95	0.71



The **Random Forest** confusion matrix (left) shows good, balanced results. Most mistakes happen between similar tree types 1 ↔ 2 and 5 ↔ 6.

The **Logistic Regression** matrix (right) has many errors, especially for small classes 4 and 5. This tells us we need a non-linear model.



Conclusion

Comparative Reflection on the Two Classification Problems

Aspect	Adult Income (Census)	Forest Cover Type
Goal	Predict if someone earns **> 50 K USD** (binary).	Predict the **tree species** (7 classes).
Rows / Features	≈ 32 000 rows, 14 original columns → 54 OHE features.	581 012 rows, 54 columns (10 continuous + 44 binary).
Imbalance	24 % positive class (moderate).	Classes 1 & 2 ≈ 85 % of data (strong).
Pre-processing	• Fill missing modes. • One-Hot Encoding for all nominal columns. • StandardScaler on numeric columns.	• No missing values. • Binary flags already 0/1, so no encoding. • Scaler used **only** for linear model; tree model kept raw values.
Feature Engineering	capital_net = capital_gain − capital_loss.	Roadway_to_Elevation and slope_ratio to mix topography with light/reachability.
Feature Selection	RFECV kept 34 / 54 features (-20 % train time).	RFECV kept 35 / 54 (-40 % train time).
Models tested	Logistic Regression (linear) and Random Forest.	Random Forest and Multinomial Logistic Regression.
Best Scores	RF Accuracy 0.82, ROC-AUC 0.915.	RF Accuracy 0.95, Macro F1 0.93.
Validation setup	80 / 20 split, 5-fold CV.	70 / 30 split, 5-fold CV.

Problem Nature

Adult Income is a **binary socio-economic task** with mixed numeric and categorical data. Forest Cover is a **multiclass ecological task** with many sparse indicator columns. The second problem is therefore both **larger** and **more complex**.

Data Representation

Because Adult data holds text categories, One-Hot Encoding was essential. In contrast, Forest data arrived in machine-friendly 0/1 flags, so encoding was skipped. This saved memory and kept the pipeline short.

Scaling Strategy

For Adult, every numeric column was scaled because *both* chosen models (linear & tree) could profit from equal ranges. For Forest, scaling was applied **only** when training Logistic Regression; Random Forest handled raw values well. This adaptive scaling reduced unnecessary work.

Feature Design

Feature ideas matched the domain:

- Adult → combine two capital columns into **capital_net** to give a clearer wealth signal.
- Forest → rate **road access** to height and mix **slope** with sunlight (**slope_ratio**) to capture vegetation conditions.

Both projects then used RFECV to cut weak variables, but kept ~⅔ of features because tree models tolerate high dimensionality.

Model Choice & Tuning

- **Adult**: Logistic Regression gave a fast, explainable baseline; Random Forest added non-linear power.
- **Forest**: A single Random Forest already fit the data well; Logistic Regression acted as a minimal baseline.

Grid-search spaces were small but adequate: four C values for LR and two depths × two n_estimators for RF in Adult; wider grids for Forest because of the bigger budget. Even with tuning, trees won both tasks, yet the *gain* was larger for the multi-class problem.

Evaluation Metrics

Adult report emphasised **ROC-AUC and per-class F-scores** to tackle a *binary* but imbalanced target. Forest report used **Macro F1** to give equal weight to every tree type and a confusion matrix to show cross-class errors. Selecting the right metric thus depended on the **label structure**.

Lessons Learned

1. **Match preprocessing to data**: encode and scale only where it helps.
2. **Tree ensembles scale** from 32 k to 580 k rows without heavy tuning and can replace many separate text-processing steps.
3. **Baseline still matters**: even when trees are likely winners, a linear model reveals if the relation is strongly non-linear.
4. **Metric must follow business need**: binary income prediction values *AUC*, while ecological mapping needs *macro* scores to protect minor classes.